

Experimental Optimization of Vortex Generators in Boundary Layer Separation Mitigation on a NACA 2412 Airfoil

Nathan Xia (student), Ekam Mann (student)

Henry Wise Wood High School, 910 75 Ave SW, Calgary, Alberta, T2V 0S6, Canada

nathanxia123@gmail.com

Abstract

Vortex generators (VGs) are passive flow control systems attached to airfoils to dampen the effects of adverse pressure gradients (APGs). VGs have already been widely implemented and optimized in commercial applications but are relatively untouched in smaller aircraft catered towards beginners, such as the Cessna 172S. This aircraft utilizes a NACA 2412 airfoil, which is a popular wing choice among smaller planes flying at low speeds. The research conducted in this paper focuses on optimizing long-gap VG spacing and VG location along the chord to reduce boundary layer separation responsible for stalling aircraft. Experiments conducted in a low-speed flow visualization wind tunnel reveal that a long gap spacing of 3.6mm at 17.2% chord length is optimal for reducing boundary layer separation at low Reynolds numbers. The optimization of these two factors proved to dramatically increase the stall angle of attack on a NACA 2412 airfoil, with an angle increase of 76% in comparison to an airfoil without VGs. This increase directly relates to stall suppression and enhances the overall aerodynamic performance of the wing. Optimizing VGs for beginner airplanes can help make flying more accessible to all.

Keywords

Engineering Mechanics, Aerospace and Aeronautical Engineering, Passive Flow Control Systems, Vortex Generators, Boundary Layer Separation.

Introduction

The biggest concern in any major industry is to lower the costs of production and services. From large corporate airlines to beginner hobby pilots, the best way to reduce costs in the aircraft industry is to increase fuel efficiency. Currently, the tuition for flight schools at top universities is extremely expensive compared to regular undergraduate programs, with some being over \$ 24,000 a year. The large entry fee of beginner programs mainly arises from the cost of maintaining airplanes, which creates a large disparity between the kinds of people with access to these programs. These costs can be cut by increasing fuel efficiency in beginner aircraft. With traditional wings reaching their maximum potential, new prospective technologies are required to improve modern aircraft fuel efficiency further. Of these technologies, both passive and active flow control systems are a potential solution to enhance aircraft aerodynamics.¹

Passive flow control systems, specifically, are methods of flow optimization that do not require external energy to operate. These techniques commonly come in the form of small protrusions across the airfoil, which affect the aerodynamic performance of the wing.² When airflow passes across the surface of the airfoil, these small protrusions manipulate the flow directly in contact with the surface, commonly referred to as the boundary layer. Flow in this layer is significantly slowed compared to free-flowing air due to frictional forces between the fluid and the wing's surface. In typical flying conditions, the boundary layer tends to stick to the surface of the airfoil throughout its entire length, which increases stability and decreases drag forces. However, as the AOA increases, the boundary layer can detach from the wing's surface, returning to the faster free-flowing air. This phenomenon is called boundary layer separation.³

Boundary layer separation is largely responsible for stalling aircraft, increasing drag, and decreasing fuel efficiency. Researchers suggest that fuel costs could be reduced by 8% if this separation could be delayed to 50%.³ This phenomenon is caused by differences in pressure across the wing, where air is

forced to flow across an unfavorable pressure gradient. On any airfoil, there is a natural pressure gradient from the leading edge to the trailing edge of the wing, as seen in Figure 1.⁴ The leading and trailing edges are typically regions of highest pressure, whereas the middle portion is typically the region of lowest pressure. As air flows from the leading edge towards the middle, it travels from a high-pressure region to a low-pressure region, creating a favorable pressure gradient. As it continues to flow, it is forced to travel from a low-pressure region to a high-pressure region, creating an unfavorable or adverse pressure gradient (APG). It is within this gradient that boundary layer separation occurs.⁵

Under normal conditions, the boundary layer contains enough energy to overcome the unfavorable gradient and continues to flow along the wing's surface. However, as the AOA increases, the adverse pressure region also does. Eventually, the boundary layer no longer contains enough energy to overcome the APG and either flows towards the region of low pressure in the free-flowing air or towards the area of low pressure in the middle of the airfoil. When the latter occurs, a recirculation region is formed where flow is reversed. In either scenario, the boundary layer is no longer attached to the airfoil and is considered to be separated. The point at which the boundary layer separates from the airfoil is the separation point, which moves up towards the leading edge as the AOA and APG increase.⁶

Vortex generators (VGs) are the most popular in mitigating boundary layer separation at high AOA of all the passive flow control techniques researched. VGs can come in many shapes and forms but are commonly described as small fins protruding across the width of the airfoil. These fins create macroscopic vortices across the wing's surface, which mix the boundary layer air with the faster-moving free-flowing air from above. This increases the velocity of the air at the boundary layer, borrowing kinetic energy from the free-flowing air that it combines with. Now, with more energy, the boundary layer can overcome the effects of the APG, postponing its separation.⁷

11 factors that influence VGs' effectiveness in mitigating boundary layer separation. They are as follows: height (h), cropped edge length (b), vane length (L), type, shape, pattern, long (D) and short (d) gaps among the vans, size, location, and inflow angle (β) (see Figure 2). The optimal value of these factors varies from airfoil to airfoil and often needs to be experimentally optimized for a specific type of wing. While VGs have already been implemented for commercial airplanes flying at high speeds, they have yet to be implemented for smaller airplanes flying at low speeds. In this study, the two factors that will be optimized for a low-speed NACA 2412 airfoil are long gaps among vanes (D) and location along the chord. All other factors will be controlled based on studies conducted on similar airfoils to the NACA 2412 such as 4415. Experimentation will also reveal the nature of unoptimized VGs, their effects on boundary layer separation, and the extent to which each factor influences VG effectiveness.

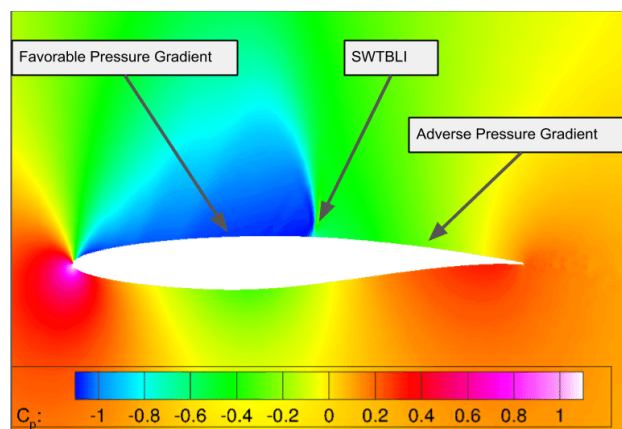


Figure 1. Natural pressure gradient around an airfoil.⁴

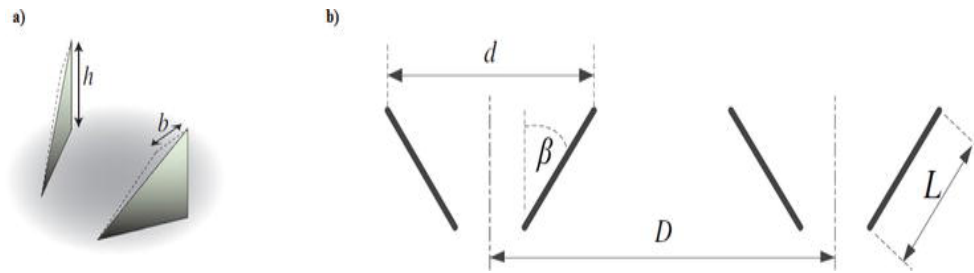


Figure 2. Various variables labeled on a VG.³

Research Question

How does the long gap width (2mm, 3mm, 4mm, 5mm) and location (10%, 15%, 20%, 25%) of VGs along a NACA 2412 airfoil affect its stall angle in low speeds (0.84m/s) at a Reynolds number below 2300?

Methods

Low-Speed Flow Visualization Wind Tunnel

The wind tunnel used in this study was constructed manually using parts at home (See Figure 3). Detailed step-by-step instructions to build a similar visualization wind tunnel can be found at nasa.gov.⁸ The required components for this wind tunnel are as follows: an airflow filter to decrease the Reynolds number and mimic the conditions of low-speed aircraft, a suction fan to mimic air flowing around an airfoil in flight (see Figure 4), a removable back plate to swap airfoils between trials (see Figure 5), an attached protractor to measure the AOA between trials, and a smoke-creating device to visualize airflow inside the wind tunnel.

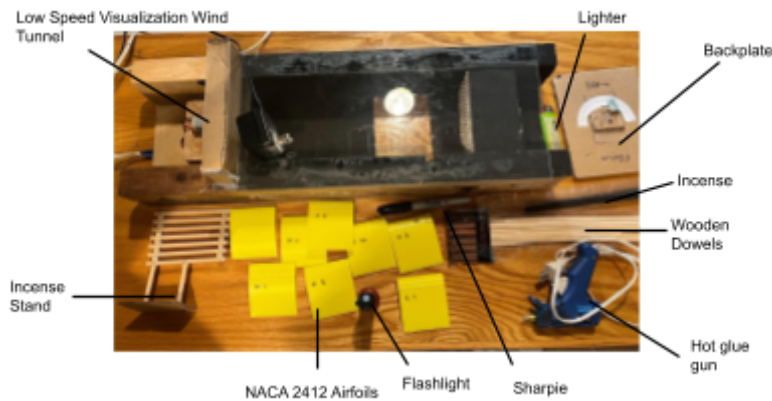


Figure 3. Apparatus diagram



Figure 4. The suction fan inside the testing chamber of the wind tunnel.

The wind tunnel used had a testing chamber that was 7.50cm wide, 34.0cm long, and 12.0cm high. It had a Reynolds number of less than 2300 (Laminar Flow), and an airspeed of 0.84m/s.

NACA 2412 Airfoils

The airfoils used in the study were mathematically designed using graphing software before being 3D printed in PLA (see Figure 6). The 2412 airfoil follows the NACA 4-digit airfoil series in the form NACA MPXX⁹.

M = Max camber that has a value of $\frac{M}{100} \times 100\%$ as a percentage of the chord

P = Location of max camber where $\frac{P}{10} \times 100\%$ is its location from the leading edge as a percentage of the chord

XX = Max thickness that has a value of $\frac{XX}{100} \times 100\%$ as a percentage of the chord

Given this, the NACA 2412 airfoil has a max camber of 2% at 40% chord length while having a max thickness of 12%.¹⁰ Assuming that the size of the chord is 1 unit, the following formulas and steps can be taken to graph a 4-digit airfoil. Note that in these formulas, both x and y refer to the x and y coordinates in a plane and are not variables.

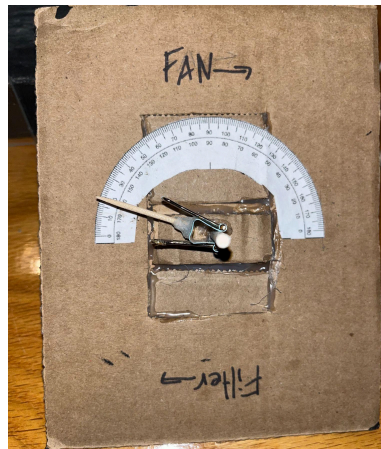


Figure 5. Removable backplate with protractor attached.

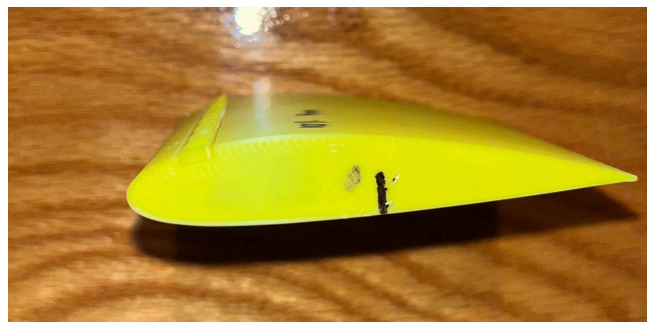


Figure 6. 3D printed PLA airfoil with VGs located at 10% chord length with 4mm spanwise spacing.

Formulas to graph mean camber line:

$$f(x) = \frac{M}{P^2}(2Px - x^2) \quad 0 \leq x < P \quad \text{(Eq. 1)} \quad f(x) = \frac{M}{(1-P)^2}(1 - 2P + 2Px - x^2) \quad P \leq x < 1 \quad \text{(Eq. 2)}$$

Formulas to determine the angle of camber:

$$a(x) = \arctan(f'(x)) \quad \text{(Eq. 3)}$$

$$\text{Where } f'(x) = \frac{2M}{P^2}(P - x) \quad 0 \leq x < P \quad f'(x) = \frac{2M}{(1-P)^2}(P - x) \quad P \leq x < 1$$

Formula to determine thickness distribution across the airfoil:

$$h(x) = \frac{XX}{20} (0.2969x^{1/2} - 0.1260x^2 + 0.2843x^3 - 0.1036x^4) \quad \text{(Eq. 4)}$$

* Note that these coefficients are constant for an airfoil that has a max thickness of 20%. This formula scales up or down based on these constants.

Formula to graph the top and bottom lines on the airfoil:

Top Line:

$$\begin{aligned} x_U &= x - h(x)\sin(a(x)) \\ y_U &= f(x) + h(x)\cos(a(x)) \quad \text{(Eq. 5)} \end{aligned}$$

Bottom Line:

$$\begin{aligned} x_L &= x + h(x)\sin(a(x)) \\ y_L &= f(x) - h(x)\cos(a(x)) \quad \text{(Eq. 6)} \end{aligned}$$

Where x_U and y_U are the coordinates that form the upper surface, and x and y are the coordinates that form the lower surface, both translated from the original x and y values.

Using these formulas in graphing software will produce a NACA 2412 airfoil which can then be converted to a .stl file to be 3D printed.¹¹ Eight airfoils scaled to 75mm by 65mm were printed, including one airfoil without vortex generators. The chord of each airfoil was 75mm. Each vortex generator was 3D modeled using TinkerCAD and separately assembled onto the airfoil after the printing.

Variables

Controlled:

- VG Height: 2mm¹²
- VG Vane length: 4.38mm (5.8% chord length)¹³
- VG Shape: Triangular¹⁴
- VG Type: Triangular Vane; Counter Rotating¹⁵
- VG Inflow angle: 12°¹⁵
- 0.84m/s airspeed
- Wind tunnel Reynolds number
- Testing chamber volume
- Boundary layer separation point
- Airfoil Type

Manipulated Variables (see Table 1 for each manipulation):

- VG Long gap spacing: 2mm, 3mm, 4mm, 5mm)
- VG Location: 10%, 15%, 20%, 25% of the chord length (75mm)

Table 1. Airfoil Specifications for Each Manipulation

Airfoil Number #	VG Specifications for Long Gap Spacing Manipulations
1	2mm @ 10% chord length
2	3mm @ 10% chord length
3	4mm @ 10% chord length*
4	5mm @ 10% chord length
	VG Specifications for Location Spacing Manipulations
3	4mm @ 10% chord length*
5	4mm @ 15% chord length
6	4mm @ 20% chord length
7	4mm @ 25% chord length
8 (Controlled)	N/A

*4mm @ 10% chord length airfoil is repeated between both manipulated variables.

The AOA of each airfoil was increased until the separation point reached 30% of the chord length from the leading edge (see Figure 7 for an example). This was observed qualitatively, and the corresponding AOA was then recorded as quantitative value. Stalling usually occurs when the separation point reaches the point of max thickness, which is located at 30% chord length on a NACA 2412.¹⁶ Furthermore, vortex generators mitigate boundary layer separation in general, and therefore choosing the separation point to record the AOA is arbitrary.

Procedure

Begin by gathering the airfoils and wind tunnel, along with all the other required materials. This includes 5-10 incense sticks, an incense stand, 8 10cm wooden dowels, a hot glue gun, a lighter, a sharpie, and a flashlight. Using the sharpie, mark out the 30% chord length point on each airfoil from the leading edge. Then, attach the wooden dowel to the marked point using the hot glue gun. Once this has been completed, insert the airfoil into the wind tunnel using the backplate and turn on the wind tunnel. It is crucial to ensure that the airfoil is initially at 0° AOA to avoid any random error. Insert the incense sticks into the stand and ignite them using the lighter before quickly blowing out the sticks to produce visible smoke. Using the flashlight to increase smoke visibility, slowly increase the angle of attack until the airflow is visibly separated from the airfoil at the marked point. You will know that the boundary layer has separated correctly when the recirculation region flows up until the marked point (see Figure 7). Once this has been achieved, record the AOA using the protractor on the backplate and then reset the airfoil to an AOA of 0°. Slowly increase the AOA again, only viewing the AOA once separation has occurred to remove unconscious bias between trials. Repeat and record the AOA for 3 trials per airfoil. Finally, repeat this entire process for all 8 airfoils, before unplugging the wind tunnel and cleaning the experiment space.

Results and Discussion

Raw Data

Table 2. Boundary layer separation AOA at 10%, 15%, 20%, and 25% VG location as a percentage of the chord. The table shows that the highest AOA before boundary layer separation occurs at 20% chord length while the lowest AOA occurs at 20%.

Airfoil Type	Trial #1 (Degrees)	Trial #2 (Degrees)	Trials #3 (Degrees)
Control	8.0	6.5	5.5
10% @ 4mm	6.0	8.0	9.0
15% @ 4mm	7.5	8.5	10.0
20% @ 4mm	13.0	9.0	12.5
25% @ 4mm	6.5	7.5	6.5

Table 3. Boundary layer separation AOA at a VG long gap spacing of 2mm, 3mm, 4mm, and 5mm. The table shows that the highest AOA before boundary layer separation occurs at 4mm spacing while the lowest AOA occurs at 2mm.

Airfoil Type	Trial #1 (Degrees)	Trial #2 (Degrees)	Trials #3 (Degrees)
10% @ 2mm	7.5	3.0	5.0
10% @ 3mm	6.0	8.5	6.0
10% @ 4mm	6.0	8.0	9.0
10% @ 5mm	4.5	7.0	5.5

Average Angle Sample Calculation

For control:

$$\theta_{avg} = \frac{Trial\ #1 + Trial\ #2 + Trial\ #3}{3} \text{ (Eq. 7)} \quad \theta_{avg} = \frac{8.0^\circ + 6.5^\circ + 5.5^\circ}{3} \quad \theta_{avg} = 6.\overline{6}^\circ = 6.7^\circ$$

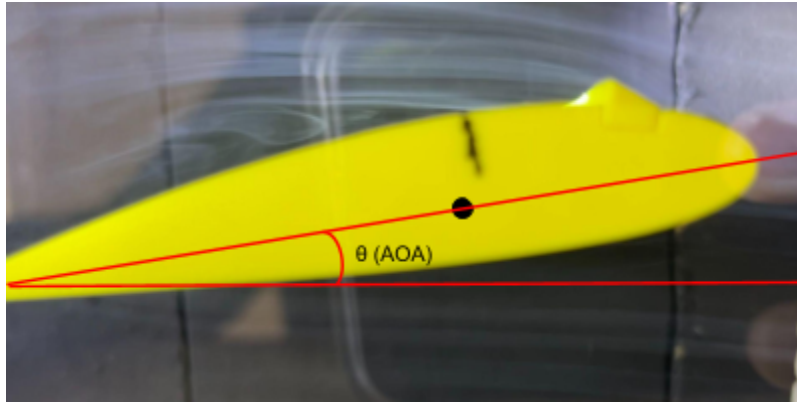


Figure 7. Boundary layer separation at 30% for airfoil #2 with θ outlined.

Processed Data

Table 4. Average AOA for boundary layer separation at 10%, 15%, 20%, and 25% VG location as a percentage of the chord. Table 4 confirms the main findings from Table 2.

Airfoil Type	Average AOA (Degrees)
Control	6.7
10% @ 4mm	7.7
15% @ 4mm	11.2
20% @ 4mm	11.5
25% @ 4mm	6.8

Table 5. Boundary layer separation AOA at a VG long gap spacing of 2mm, 3mm, 4mm, and 5mm. Table 5 confirms the main findings from Table 3.

Airfoil Type	Average AOA (Degrees)
10% @ 2mm	5.2
10% @ 3mm	6.8
10% @ 4mm	7.7
10% @ 5mm	5.7

Graphs

Average Boundary Layer Separation Angle as a Function of Vortex Generator Placement

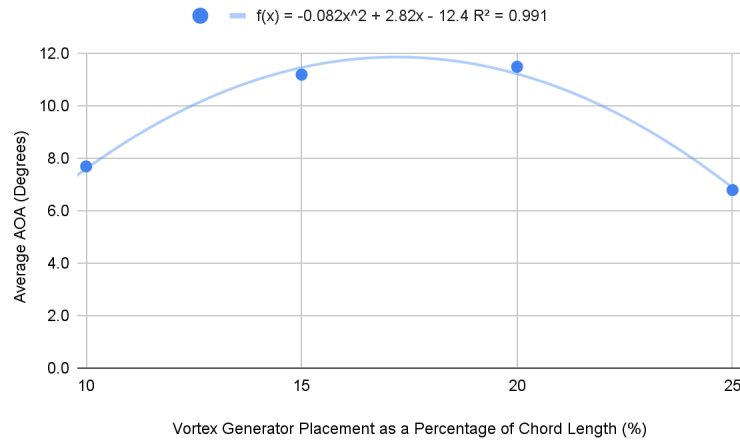


Figure 8. Average boundary layer separation angle as a function of VG location. The graph shows that the optimal location for VGs is located between 15% and 20% as indicated by the maxima. This produces an AOA just below 12.0 degrees.

Average Boundary Layer Separation Angle as a Function of Long Gap Spacing

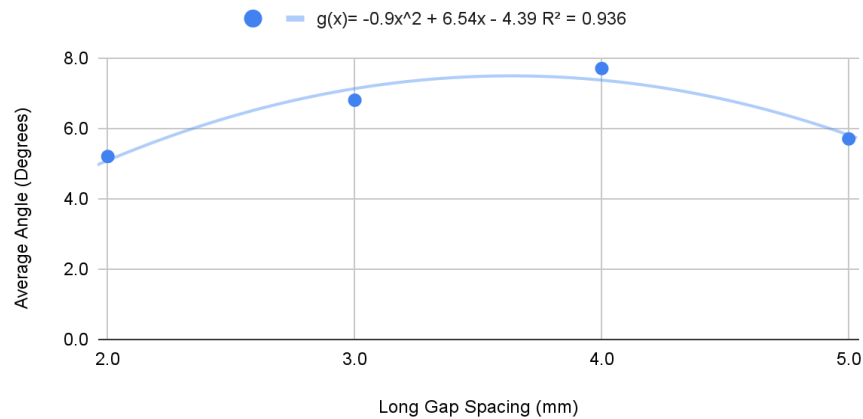


Figure 9. Average boundary layer separation angle as a function of VG long gap spacing. The graph shows that the optimal long gap spacing for VGs is between 3 and 4 mm as indicated by the maxima. This produces an AOA of between 6.0 and 8.0 degrees.

Maximum Curve Calculations

For Figure 8:

$$f'(x) = 2(-0.082)x + 2.82$$

$$f'(x) = -0.164x + 2.82$$

$$0 = -0.164x + 2.82$$

$$0.164x = 2.82$$

$$x = 17.\overline{19512} = 17.2\%$$

$$y = (-0.082)(17.\overline{19512})^2 + (2.82)(17.\overline{19512}) - 12.4 = 11.845\dots$$

$$y = 11.8^\circ$$

For Figure 9:

$$g'(x) = 2(-0.9)x + 6.54$$

$$g'(x) = -1.8x + 6.54$$

$$0 = -1.8x + 6.54$$

$$1.8x = 6.54$$

$$x = 3.\overline{63} = 3.6\text{mm}$$

$$y = (-0.9)(3.\overline{63})^2 + (6.54)(3.\overline{63}) - 4.39 = 7.491$$

$$y = 7.5^\circ$$

The experiment was intended to optimize the placement and long gap spacing of vortex generators on the NACA 2412 airfoil, aiming to mitigate boundary layer separation, which would increase aerodynamic efficiency and postpone stalls. Eight airfoils with varying VG configurations concerning long gap spacing and location on the chord were 3D-printed. The AOA on each airfoil was increased until the boundary layer separation reached 30% of the chord length from the leading edge.

Table 4 summarizes the AOA values for different VG locations across the chord, given a fixed long gap spacing at 4mm. Table 4 was derived from the raw data in Table 2 using Equation 7. The highest AOA of 11.5° was observed at placement at 20% of the chord, and the lowest value, 6.8° , occurred at 25%. This indicated that placing VGs too close to the trailing edge reduced effectiveness in delaying boundary layer separation, likely because of insufficient interaction between the boundary layer and free-stream air at higher chord lengths. Oppositely, the 20% chord position gave the most efficient boundary layer and free-stream air combination, allowing for better flow attachment at higher AOAs. Long gap spacing and its effects are explored in Table 5. Table 5 was derived from the raw data in Table 3 using Equation 7. 2mm, the smallest gap spacing resulted in the lowest average AOA (5.2°). A gap spacing of 4mm produced the highest average AOA. We interpret this as meaning that having more vortices isn't always better, as the reduced gap spacing might actually reduce their strength. This would, in turn, lead to a less effective mixing of boundary layer and free-stream air. The vortices generated by the 2mm VGs may begin to interfere destructively with the ones generated destructively, leading to less effective vortices overall. Furthermore, the increase in VGs could lead to an increased parasitic drag. Adding VGs increases the amount of drag created due to form resistance, which also increases the tendency for airflow to separate. This effect is generally outweighed by the positive effects of properly optimized VGs (at 3mm and 4mm). Still, it tends to stick out in poorly optimized VGs with negligible positive effects.

A parabolic regression line was used to model both tables of values, each with an R^2 value of above 0.936. Finding the maximum value of both these equations gives the maximum angle possible for each manipulation and the exact value for this max. In Figure 8, the maximum value reached is 11.8° at a VG location of 17.2% from the airfoil's leading edge. This value represents a 76% increase in angle from the controlled airfoil. In Figure 9, the maximum value reached is 7.5° at a VG long gap spacing of 3.6mm. This value represents a 12% increase in angle from the controlled airfoil. Given this, it is evident that the location of the VGs plays a much greater role in postponing stalls in aircraft than the long gap spacing of the VGs. This could indicate that positioning the vortices along the chord is more effective in mitigating boundary layer separation than the size of the vortices generated.

Conclusion

Modern airfoil designs have reached their maximum potential in aerodynamic performance to keep up with the constant need for increased fuel efficiency. Given this, innovative and modern solutions are required to improve aircraft performance further. Of these, passive flow control systems such as vortex generators show great potential as a solution to aerodynamic performance, significantly postponing stalls and boundary layer separation. The effectiveness of these generators is defined by 11 factors that need to be specifically optimized for each airfoil and flying condition.

Our study optimizes two of these 11 factors on a NACA 2412 airfoil, targeting beginner aircraft flying at low Reynolds numbers and airspeeds to help reduce the costs of entry. These factors include the long gap spacing between VG sets and their location on the chord. These factors were optimized experimentally in a low-speed visualization wind tunnel based on observations of flow separation at 30% of the chord. The results from this study indicate that a long gap spacing of 3.6mm at 17.2% chord length is optimal for the NACA 2412, showing a significant increase in AOA before flow separation.

Experimentation results show that on a blank airfoil, the separation point reaches 30% chord length at an AOA of 6.7° , whereas optimized VGs delay the AOA up to 11.8° . Furthermore, the results of this study indicate that the location along the airfoil plays a much greater role in delaying flow separation than the long gap spacing, as the long gap spacing alone only delayed the AOA to 7.5° before the separation point reached 30%. Something else to consider is that improper implementation of VGs can increase flow separation and decrease the AOA for stalls compared to a blank airfoil. This is seen in VGs with 2mm and 5mm spacings, as their AOA for separation is recorded as 5.2° and 5.7° , respectively. This performance is significantly worse than the blank airfoil at 6.7° and is most likely the result of parasitic drag.

Further improvements and research include measuring each airfoil's lift and drag forces and determining the ratio of their coefficients. Determining how the lift and drag coefficients are affected by the location of VGs on a NACA 2412 airfoil would provide valuable information and insight into their effects on parasitic drag as well. Another area of further research could be optimizing other factors, including intake angle or size.

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Authors

Nathan Xia is a Senior at Henry Wise Wood High School in Calgary, AB. With interests in physics and aerodynamics, he plans to continue his passions in university through a mechanical or aerospace engineering degree. Nathan hopes to attend the University of Toronto in the fall of 2025.

Ekam Mann is a Senior at Henry Wise Wood High School in Calgary, AB. He is a current air cadet who's spent time learning from TOPGUN graduates and industry professionals at US Air Force installations. Ekam hopes to earn an aerospace engineering degree and pursue a career as a pilot.